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A Milk-run routing and Scheduling model for a Smart Manufacturing System

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Abstract: The fourth industrial transformation can create value for companies. It can bring significant improvements: those who want to be competitive in the market must implement new methods and tools to make plants efficient and productive. Industry 4.0 introduces new paradigms promoting the adoption of innovative technologies. In this context, the material handling system plays a crucial role in increasing the effectiveness and efficiency of the manufacturing systems. This paper proposes a model based on the Vehicle Routing Problem with Time Windows. The model aims to plan a dynamic routing strategy for milk-run to deliver the materials to different working stations, ensuring a materials' distribution consistent with the time slot and minimizing the tugger trains path. A real industrial full case of a worldwide automotive company validates the model. The results of the study show benefits under economic (cost saving of 18%), environmental (allowed to reduce the distance travelled by tugger trains by around 23% for each work shift) and social (organizational development) perspectives.

Keywords: Industry 4.0, Smart manufacturing systems, material handling systems, in-plant milk run, milk-run routing, vehicle routing problem, time window constraints.

1. INTRODUCTION

The latest stage in today's industrial revolutions, so-called Industry 4.0, continues to bring significant change in manufacturing, consumption, and supply processes from an economic and environmental perspective (S. Digiesi, Mossa, & Rubino, 2015). The new approach consists of machines and production systems that do not require manpower and operate independently of people (Erboz, 2020). The potential of new devices ensures a real-time reaction to changes and quality response in satisfying the customer needs of manufacturing companies (Azizi et al., 2015). Consistently with the new approach, the 'smart factory' should be enhanced by implementing new additional features providing a flexible and adaptive production process. Therefore, to make a factory 'smart', it is needed to implement new data management, share and monitor all process information, and minimize unnecessary labour and waste of resources (Radziwon, Bilberg, Bogers, & Madsen, 2014).

In this context, the material handling system (MHS) plays a crucial role in increasing the effectiveness and efficiency of the manufacturing process (Mathew & Sahu, 2018). According to Zhang, Lv, & Zhang (2018), material handling costs accounted for 20–50% of total manufacturing costs. Therefore, developing effective and efficient handling strategies has received considerable attention over the last few years. This has led to rapid development in MHS design, allowing to scale the production and increase the developed systems' accuracy and adaptability. A 'smart' MHS is developed to improve the communication between units in the system to create more adaptable control of assembly flow and system performance (Azizi et al., 2015). There are many approaches to operate in-plant logistics systems (Salvatore Digiesi, Mascolo, Mossa, & Mummolo, 2016). One of increasing interest, especially for lean production systems with small lot sizes, is referred to as a milk-run method (Greenwood, Kluska, & Pawlewski, 2017). It allows for increasing the frequency of materials delivered, ensuring the supply of working stations, and minimizing the materials stocked. Based on the milk-run approach, the tugger trains have become a popular means of supply in material handling intensive production systems (G. Bocewicz, Nielsen, & Banaszak, 2019). Generally, they are adopted in supermarkets to interlink multiple delivery locations by adopting a transport route designed according to the milk-run method (Grzegorz Bocewicz, Bozejko, Wójcik, & Banaszak, 2019). Tugger trains are loaded and unloaded multiple times along a route that connects several working stations into a supply loop. This setup ensures the delivery of a variable quantity of materials to a given set of working stations.

On the one hand, the milk-run method ensures an efficiency gain in terms of higher transport capacity and reduced labour costs. On the other hand, the method requires a complex design due to complicated planning and a high number of variables to be considered. Nevertheless, the operational benefits provided by the milk-run method are unquestionable, especially in the automotive sector, where many different materials are required, and the area for the material provision (very close to the assembly line) is limited. Consistently with the lean manufacturing perspective, the milk-run approach reduces the cost due to waste from inventory, delivering the material in

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time and minimizing, at the same time, the waste in processing, avoiding the working stations' starvation (Alnahhal, Ridwan, & Noche, 2014). Therefore, the milk-run routing and scheduling problem consists of planning in what time windows a batch of specific materials shall be delivered to specific working stations (Grzegorz Bocewicz et al., 2021).

In the general case, the research to identify the optimal distribution policy evaluating a set of routes ensuring the delivery in a specific time slot can belong to the class of Vehicle Routing Problem (VRP). New challenges in particular concern the development of models allowing to solve the milk-run-driven delivery problems under specific constraints are under investigation. In practice, these kinds of assumptions on a shop floor include a stochastic demand of materials over time to be managed through a dynamic plan of routing (developed online) to minimize various objectives such as cost, distance, or time, adopting I4.0 digital tools.

Therefore, the paper contributes to the practice and methodology by providing a model based on the Vehicle Routing Problem with Time Windows (VRPTW). The model aims to plan a dynamic routing strategy for milk-run to deliver the materials to different working stations, ensuring a materials' distribution consistent with the time slot and minimizing the tugger trains path. It was assumed that the vehicles are dynamically dispatched from the depot to working stations when the materials' orders are processed. The main contributions of this paper can be summarized as follows: the materials will be moved along variable routes at changeable time intervals (i); the effectiveness of the model is tested on a real full case study of an automotive company (ii); the model is implemented in a smart factory adopting I4.0 technologies (iii).

The remainder of this paper is organized as follows. Section 2 is about the literature review of using in-plant milk run and previous studies about VRPTW. The model is described and tested in an automotive company, in sections 3 and 4, respectively. Finally, section 5 highlights the conclusions, contributions of the work, limitations, and future research suggestions.

2. LITERATURE REVIEW

The literature discusses two categories of milk-run systems: the external and the internal (in-plant) milk-run logistic system. The first category (also so-called downstream milk runs) includes the transport of parts from the manufacturing plant to its customers or vice versa. The second one (also so-called upstream milk runs) manages the inbound delivery network to the manufacturing plant (Kodippili & Samarasekera, 2019). Most studies addressing outbound milk-run systems focus on the routings and schedules of the vehicle fleet used. Generally, they adopt mathematical models based on heuristic approaches using different metaheuristics, such as hybrid ant colony optimization and Tabu search (Grzegorz Bocewicz, Nielsen, Jasiulewicz-Kaczmarek, & Banaszak, 2020). Similar approaches identified to the milk-run systems outside the plant are adopted in the milk-run systems inside the plant. In this case, the milk-run distribution problem identifies the routes and time periods of routes under the constraints such as space requirements, number of vehicles, the capacity of vehicles, etc. According to Kilic, Durmusoglu, & Baskak (2012), route and time are two critical factors for the in-plant milk-run problem categorization. Consistent with these factors, three classes of the problem can be identified:

- Generalized assignment problem: routes and time periods are not known.
- Dedicated assignment problem: the routes are known; the problem consists of identifying the time periods.
- Determined time periods assignment problem: time periods are known, but the routes are unknown.

The VRP is the most common approach adopted to identify, for each category, the set of routes compatible with the layout of the manufacturing plant and the product flow. Different constraints lead to a variety of task-specific problems depending on the specific characteristics of the problem. In this regard, Grzegorz Bocewicz et al. (2021) identify two clusters of the problems defined as "relatively simple", including Mix Fleet VRP, Multi-depot VRP, Split-up Delivery VRP, Pick-up and Delivery VRP, or VRP with Backhauls. The second cluster, identified as "complex problems", includes VRP with multi-trip multi-traffic pick-up and delivery with time windows and synchronization, which is a combination of variants of the VRP with multiple trips, VRP with a time window, and VRP with pick-up delivery. Due to the continuing developments of the I4.0 technologies, vehicle routing and scheduling problems have received special research attention in the last years (Adriano, Montez, Novaes, & Wangham, 2020). Most of the research included in the first cluster is devoted to the analysis of methods of organizing transport processes in ways that minimize the size of the fleet, the distance travelled, or the space occupied by a distribution system. A mixed-integer linear programming model was developed to minimize the fluctuations in the number of milk-run vehicles between workdays and production facilities. The model achieved the target of the study by identifying the minimal routes, increasing the loading rate of the milk-run vehicles, and avoiding the coincidence of the milk-run at the depots (Aksoy & Öztürk, 2016). A model for a milk run pickup problem on the distribution of automobile parts with uncertain demand and frequency, considering the balance between inventory cost and distribution, was developed in 2018. The research identifies two best solutions: one leads to increased schedule stability, and one minimizes the schedule cost. According to the authors, the decision-maker could prefer a stable schedule for the sale season and a low-cost schedule for the slow season (Wu, Wang, He, Xuan, & He, 2018). In the second cluster, there are 'complex problems', which include a combination of variants of the VRP with multiple trips, VRP with pick-up delivery, and VRPTW, among others. In this regard, an empirical analysis for the VRPTW with a multigraph representation of the road network was conducted by Ticha, Absi, Feillet, & Quilliot (2017). The authors developed a simple multigraph representation in this research work to significantly improve the VRPTW solution identified by adopting a Branch-and-Price algorithm. Similarly, a Multi-objective VRPTW was solved by adopting a tissue P system with three cells providing a set of Pareto solutions (instead of a single solution) in a short time. In this way, the decision-maker knows sufficient choices to make the best decision (Dong, Zhou, Qi, He, & Zhang, 2018). The research work belongs to this cluster of problems; the materials will be moved along variable routes at changeable time intervals.

3. MODEL DEVELOPMENT

3.1 List of notations

The following notations are adopted to describe the mathematical model developed:
[i, j]: starting and finishing node,
n: number of nodes/customers,
m: number of vehicles of the fleet,
D: capacity of a vehicle,
$T_{k,\, route}$: allowed maximum time to complete the route at k-vehicle,
d_j: demand of the node/customer j to be served,
$t_{k,j}$: the time required by k-vehicle to load/unload the materials demanded by the customer j, independently of routing travelled,
$t_{k,ij}$: time needed to k-vehicle to travel from node i to node j,
c_{ij}: distance to travel from node i to node j,
$x_{k,ij}$: it is a binary variable, corresponding to 1 when the k-vehicle travels from node i to node j, else it is 0,
$x_{k,i}$, $x_{k,j}$: they are binary variables, corresponding to 1 when the k-vehicle is in node i or j, respectively, else it is 0,
[a_i, b_i]: represents the interval of the time window,
$s_{k,i}$, $s_{k,j}$: the timestamps when the k-vehicle leaves the nodes i and j, respectively.

3.2 Model formulation

The mathematical formulation of the VRPTW is based on an algorithm for a vehicle routing problem with realistic constraints (D. Zhang, Cai, Ye, Si, & Nguyen, 2017). The problem defines a binary variable $x_{k,ij}$ equal to 1 if the k-vehicle follows the arc of the graph *(i, j);* otherwise, it is 0. Generally, the models to describe transport problems use graphs. A general graph is named $G = (V, E)$, where $V = \{V_i\ V_n\}$ represents a set of nodes which model the customers (or cities/locations) and $E = \{(Vi,Vj): i\leq j\}$ represents a set of arcs connecting the customers. Overall, the VRPTW is part of the class of optimization problems (Dridi, Mailhot, Parizeau, & Villeneuve, 2009).

The VRPTW's mathematical formulation aims at minimizing the total distance of the routes, identified as cost function (eq. 1), while respecting the framework to perform the delivery on time in the time window constraints.

$$Cost(x) = Min \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{m} c_{ij} x_{k,ij} \quad (1)$$

Further objectives could be considered, for instance, minimizing the penalties occurring when some constraints are violated, such as the time windows constraints. Another target consists of using the minimum number of vehicles in the fleet to perform all the deliveries at the minimum cost. In general, these two objectives are in contrast; therefore, solving the optimization problem can provide the optimal compromise to schedule the deliveries.

The function to minimize (eq. 1) is subject to the following constraints:

$$\sum_{i=1}^{n} \sum_{k=1}^{m} x_{k,ij} = 1\ \forall 1 \leq j \leq n \quad (2)$$

$$\sum_{i=1}^{n} \sum_{k=1}^{m} x_{k,ij} = 1\ \forall 1 \leq i \leq n \quad (3)$$

$$\sum_{j=2}^{n} x_{k,1j} \leq 1,\ k = 1, \ldots, m \quad (4)$$

$$\sum_{i=2}^{n} x_{k,i1} \leq 1,\ k = 1, \ldots m \quad (5)$$

Equations 2 and 3 ensure that each customer is served only one time and only by one vehicle. The constraints expressed with equations 4 and 5, respectively, show that each vehicle leaves the depot and returns to it only once.

$$\sum_{i=1}^{n} x_{k,i} - \sum_{j=1}^{n} x_{k,j} = 0,\ k = 1, \ldots, m; \quad (6)$$

Equation 6 establishes that when a node is visited, it must imperatively be left (after materials' delivery), as the route traversed by a vehicle must be continuous.

$$\sum_{j=1}^{n} t_{k,j} \sum_{j=1}^{n} x_{k,j} + \sum_{i=1}^{n} \sum_{j=1}^{n} t_{k,ij} x_{k,ij} \leq T_{k,route},\ k = 1, \ldots, m \quad (7)$$

As ensured by equation 7, the total duration of a route must be respected. Similarly, each vehicle's capacity must be respected, as shown in equation 8.

$$\sum_{i=1}^{n} d_j \left(\sum_{j=1}^{n} x_{k,j}\right) \leq D, k = 1, \ldots m \quad (8)$$

$$x_{k,ij}\left(s_{k,i} + t_{k,ij} - s_{k,j}\right) \leq 0 \quad (9)$$

$$a_i \leq s_{k,i} \leq b_i \quad (10)$$

Finally, equations 9 and 10 ensure that the temporal constraints, indicated by the time windows, are respected.

3.3 Model implementation

The VRPTW has been implemented in the Python language. Stated that the model concerns the optimization of the routes of tugger trains, each of them composed of three wagons and used to deliver pallets to the working stations of a shop floor; Figure 1 represents the model's flow chart. After the parameter initialization, the first task of the algorithm consists of reading the timestamp corresponding to each wagon of the tugger trains, namely the timestamp in which a tugger driver books the pallet carried on that specific wagon. The materials were loaded on tugger trains' wagons according to an order assigned by an upstream process depending on different aspects (e.g., work stations' positions, materials' priority, frequency of delivery, etc.). The algorithm clusters the trains by checking if there are identical timestamps among all pallets. Therefore, for each train, the time windows of each wagon are calculated. For this purpose, it is adopted a matrix of time windows. It is updates in real-time by considering when the order has been sent and when the tugger driver has booked the corresponding pallet.

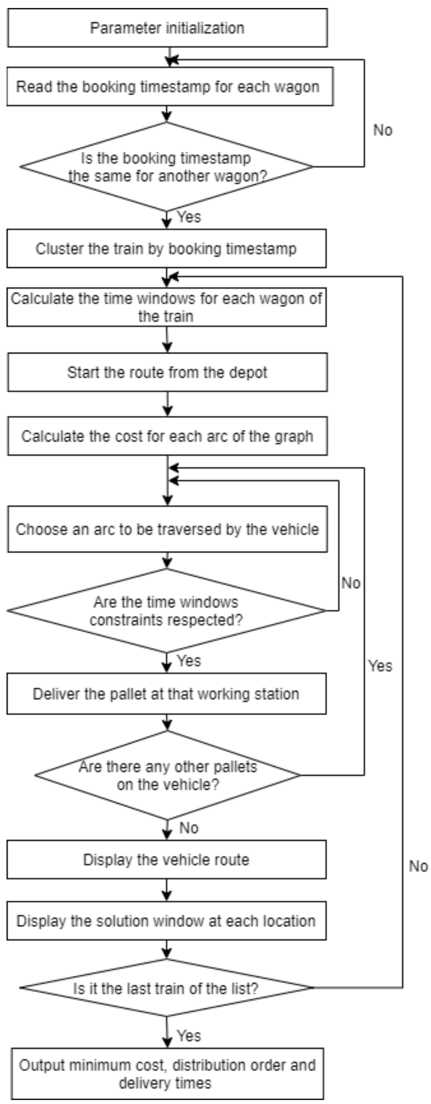

Figure 1. Flow chart of the VRPTW model

Consequently, the driver receives the information on a tablet and starts the route; instead of deciding autonomously which path to take and which pallet to deliver first, the algorithm calculates the cost for each arc of the graph according to the mathematical formulation introduced in the previous section. As a result, checking the respect of the time windows iteratively and of the model's constraints, each pallet is delivered at the corresponding working station until all the pallets of the train examined are consigned. A GPS device constantly tracks all positions of each tugger train; the data are sent to the server in real-time to update the timestamp. When a delivery is concluded, the working station sends a confirmation by RFID sensors. To conclude, the algorithm analyses the vehicle routes of all the records and the relative time window in which the delivery has been completed. The average running time is around 5 seconds per routing identification, adopting a MacBook equipped with a 1.6 GHz Intel Core i5 CPU and 4 293 GB RAM.

4. THE REAL-FULL CASE STUDY

The model presented in this paper has been applied to study a real-full case of a worldwide automotive company to optimize the internal logistics of one of its shop floors. Although the model allows managing multiple vehicles, in the case study introduced, the fleet of Material Handling Equipment (MHE) includes four tugger trains.

The issue to be tackled is how to deliver pallets containing materials to work on time at the working stations of a welding shop, in a process that involves four tugger trains of three wagons each, driven by tugger drivers assuming around 400 trips per work shift. The materials orders are dynamically sent to MHS consistent with working station needs. In this case, it is not possible to plan the demand of materials since there are high variable set-up times for each task and work station; moreover, most tasks are manually oriented; therefore, the cycle times depend on human performance and task hardness. The adopted technology involves the use of one tablet with a GPS and Wi-Fi sensor and one RFID transponder for each tugger train. Similarly, each working station is linked to a server adopting an illuminated push-button with contact blocks and one RFID tag. The traditional (no optimized) process is based on the human decision making of the drivers, who decide the routes to follow for the deliveries. Therefore, in the traditional process, the pallets are subject to late and early deliveries, which implies that the lead time is irregular, as shown in Figure 2.

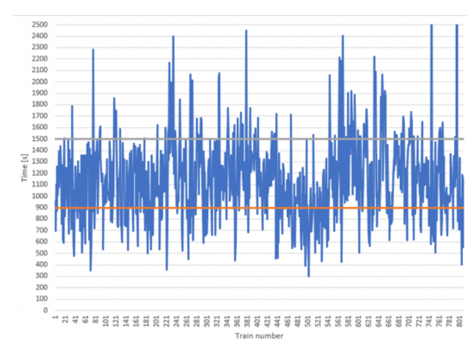

Figure 2. The lead time of the current process

According to the constraints of the production line, a pallet delivery has been categorised as 'late delivery' if more than 25 minutes have passed from when the order has been sent to when the pallet has reached the working station. On the contrary, an 'early delivery' occurs when the pallet is delivered within 15 minutes from its request. Consequently, setting the time windows constraints according to this definition of early and late deliveries, the VRPTW model was implemented, adopting devices available on the market for each tugger train, each working station, and IT infrastructure to link and manage the data flow. In terms of cost, have been estimated investment of around 5 k€ for each tagger train and about 50 k€ for the overall IT infrastructure implementation (including expenditure for hardware and software development); the cost for each working station can be considered negligible (around 0.1% of the total cost).

The results achieved by comparing the traditional process to the optimised process, show that by applying the innovative model, the number of late and early deliveries significantly decreases, and the process's lead time curve becomes more even. The curve of the lead time of the optimised process is shown in Figure 3.

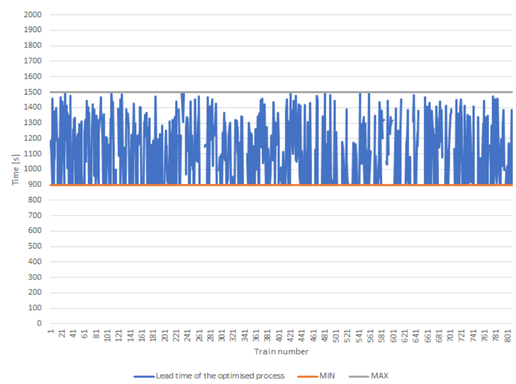

Figure 3. The lead time of the process after the VRPTW algorithm implementation

The VRPTW algorithm led to significant improvement of the delivery process, suggesting to tugger drivers the best routes to follow and eliminating the problem of the human decision making on the shop floor while increasing the number of on time deliveries. The comparison between the lead times of the process estimated before (Fig. 2) and after (Fig. 3) the implementation of the model shows that 11% of late deliveries in the current process becomes 8% in the optimized process, as indicated in Figure 4, evaluated before and after the model implementation.

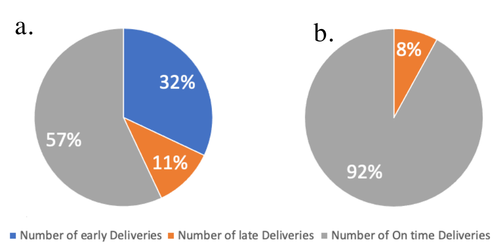

Figure 4. Number of "early", "late", and "on time" deliveries before (a) and after (b) the model implementation

Considering the above, as the tugger drivers know in advance when and where they must deliver the pallets, early deliveries do not occur in the optimised process, which increases the on-time deliveries percentage from 57% to 92%. In economic terms, there are cost-saving due to re-handling, performance, and energy costs. The optimized model reduces to zero the early delivery, avoiding the operations due to re-handling of materials delivered too early and brought back. Similarly, the late deliveries are reduced by 3%, improving the line performance and avoiding the process starvation. The route optimization impacts the electrical cost required by tugger trains charging. A rough estimate led to evaluating an overall cost reduction of around 18% compared to traditional process costs. From the environmental perspective, the path optimization reduced the distance travelled by tugger trains by around 23% for each work shift. Therefore, it was estimated an energy consumption save of about 0.9 kWh/h per work shift. The optimized process contributed to developing the workforce organization. The new process provides a clear view of the contribution of each worker to the performance of the line, reduces the no value-added tasks, and improves the communication between the tugger drivers and other workers.

6. CONCLUSIONS

The novelty of this study is that it proposes an integrated modelling approach to milk-run system design and operation, which considers the relationships linking the production flow with the replenishment batches and the time window. According to our evaluation, the model implementation led to a significant reduction in early and late deliveries (around 35%) compared to the traditional process. The investment is relatively 'low' (about 70 k€), and in most cases, the existing technologies can be upgraded to interface with the VRPTW model. We can summarize the benefits from an economic perspective, a cost saving of 18% was estimated comparing the cost of the optimized and traditional process. From an environmental perspective, the optimized routes reduced the distance travelled by tugger trains by around 23%, leading to better deployment of energy resources. Organizational development is expected due to greater clarity in communications and better management of the orders and deliveries; this aspect will benefit the workers' social life assessment and well-being. Nevertheless, the model can be further improved, ensuring higher reliability of the deliveries (8% of the materials are delivered too late) and assigning a priority scale to some working stations. In this regard, future works should include a more reliable data dissemination protocol with various milk-run operations. The optimization conducted should not be seen as 'stand-alone' but integrated into a more complex manufacturing system. In this case, the information on working stations' needs, materials inventory, tugger trains availability, etc., should be managed according to the end-to-end approach.

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